

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

8-layer 20×20mm FPD-Link III automotive camera module PCB (image sensor + serializer + power)

Populated PCB (PCBA) · from photo analysis · Built in: China (Shenzhen / Suzhou) · Delivered: UK, duty paid · Currency: GBP

Ex-works / board (China)	Delivered UK / board	Components	Bare board	Assembly & test
£14.72	£15.26	87.4%	2.7%	5.4%

Estimate Confidence: **Low** · likely £12.22 – £19.77 · 11 BOM lines, 1 to verify with a quote

Key Assumptions

Populated board built in China (Shenzhen / Suzhou), delivered to the UK, import duty paid, at 250,000 boards a year.

Ex-works £14.72 is the board built, tested and packed at the China factory gate; delivered £15.26 adds freight to the UK and UK import duty (£0.54).

Board: Automotive camera board · costed at ASIL-B · parts list from the BOM image (11 lines, 80 parts).

Component prices follow the parts bought (qty per board × boards); fabricator and EMS prices carry their own overhead and margin — none is added on top.

§1 — Cost Breakdown per Board

China (Shenzhen / Suzhou) · delivered UK, duty paid

Element	Amount	% of total	Basis
Components (as the EMS buys them)	£12.70	83.2%	11 BOM lines at China (Shenzhen / Suzhou) sourcing — see §3
EMS material burden (5%)	£0.64	4.2%	Procurement, inventory, scrap and margin on material — stated rate by volume
Bare PCB fabrication	£0.27	1.8%	Fabricator price: base, layers, finish, vias, set-up — see §4
Automotive fab grade	£0.14	0.9%	IATF 16949 line, automotive laminate, class 3 inspection, coupons
SMT / THT assembly	£0.26	1.7%	Placements and joints at the country line rate
Test & inspection	£0.39	2.6%	AOI, X-ray, ICT (station time + fixture over the order)
Automotive assembly grade	£0.18	1.2%	IATF line, class 3 workmanship, serialisation
Energy, packaging, cost of quality	£0.13	0.9%	Energy £0.01 · ESD packaging £0.06 · rework/scrap £0.06
Rounding	£0.01	0.1%	Each figure is rounded to the penny on the server

Element	Amount	% of total	Basis
Ex-works cost (China factory gate)	£14.72	96.5%	Built, tested and packed — before freight and import duty
Freight to the UK	£0.00	0.0%	Sea at volume, air below 2,500 boards — under a penny a board by sea at this size and volume
UK import duty	£0.54	3.5%	On components + board + assembly (customs value)
Delivered cost per board	£15.26	100.0%	China (Shenzhen / Suzhou) build, delivered UK, duty paid

Components, bare board and assembly are supplier prices: the fabricator's and the EMS's overhead and margin are inside them, so no further overhead or margin is added.

§2 — Basis & Confidence

Build country	China (Shenzhen / Suzhou)	Annual volume	250,000 boards
Delivery	UK, import duty paid	Board type	Automotive camera board
Parts list from	BOM image	Lines · parts	11 lines · 80 parts
Estimate confidence	Low	Likely range	£12.22 – £19.77
Lines to verify	1 (£4.64)	Meaning	£1.00+ lines with no distributor price

Functional Safety (ISO 26262)

Costed as ASIL-B (the photo classifier said ASIL-C); quality grade auto grade 2. An ASIL comes from the hazard analysis (HARA), not from a photograph — confirm it against the safety concept.

The classifier described a radar module, but the parts list is a camera board — its description is withheld as inconsistent with the BOM.

ASIL-C was claimed from the photos, but the parts list has no safety PMIC / SBC or lockstep safety MCU that an ASIL-C design carries — costed at ASIL-B (no burn-in). Confirm the level against the safety concept; an ASIL-C answer adds burn-in and the higher NRE tier.

§3 — Bill of Materials

11 lines · 80 parts · £12.70

#	Description / part number	Package	Qty	Unit	Ext	Priced from
1	Ceramic Capacitor	<= 1206	49	£0.02	£0.91	class range
2	Ferrite		7	£0.04	£0.30	class range
3	Common Mode Choke		1	£0.24	£0.24	class range
4	Common Mode Choke		1	£0.24	£0.24	class range
5	FPD-Link III Serializer With CSI-2 Interface DS90UB935-Q1	VQFN-32	1	£1.87	£1.87	catalogue (distributor)
6	Automotive Step Down DC-DC Converter LM53600MQDSXRQ1	WSOP-10	1	£1.31	£1.31	catalogue (distributor)
7	2.25-MHz Fixed VOUT Dual 1000mA/400mA Step-Down TPS62422QDRCRQ1	MSOP-10	1	£0.82	£0.82	catalogue (distributor)
8	300-mA, Low-IQ, Low-Dropout Regulator TLV70233QDSERQ1	WSOP-6	1	£0.17	£0.17	catalogue (distributor)

#	Description / part number	Package	Qty	Unit	Ext	Priced from
9	LI-ION AND LI-POL CHARGER IC BQ24025DRCR	VSON(DRC)10	1	£2.10	£2.10	catalogue (distributor)
10	Resistor	<= 1206	16	£0.01	£0.08	class range
11	Image sensor		1	£4.64	£4.64	class range *
COMPONENTS TOTAL			80		£12.70	

The 11 lines (80 parts) total £12.70 — the components row of §1. Unit prices are the distributor price × 0.879, China (Shenzhen / Suzhou)'s component sourcing index. The EMS material burden (£0.64) is added on top in §1.

Source: catalogue (distributor) = a distributor price on file; catalogue (estimate) = a dated engineering estimate; named-part range = the part is identified, priced inside its own range; function / class range = priced from the tool's table for that kind of part. Lines marked * are £1.00+ with no distributor price behind them — confirm with a quote.

§4 — Board & Assembly

Size	20 × 20 mm (estimated from the photos)
Layers	8
Technology / HDI	FR4 HTg
Finish · copper	OSP · 2 oz
Vias (through / blind / micro)	115 / 0 / 0
Impedance control	Yes
Placements · BGA/CSP	80 SMT · 1 BGA/CSP · 2 reflow side(s)
Panel	280 boards on 480 × 350 mm (67% used)
Bare-board build-up	base £0.01 · layers £0.03 · vias £0.22 · copper £0.01 · impedance £0.01

§5 — Build-Country Comparison

each row: this board built there — ex-works, and delivered UK duty paid

Country	Components	Bare board	Assembly	Other	Ex-works	Freight + duty	Delivered	vs China	Lead
China	£13.34	£0.41	£0.83	£0.14	£14.72	£0.54	£15.26	costed	3 wk
Vietnam	£13.95	£0.49	£0.52	£0.13	£15.09	£0.83	£15.92	+£0.66	3 wk
Malaysia	£13.80	£0.57	£0.92	£0.13	£15.42	£0.84	£16.26	+£1.00	3 wk
Thailand	£14.10	£0.58	£0.79	£0.15	£15.62	£0.85	£16.47	+£1.21	3 wk
India	£14.40	£0.52	£0.73	£0.17	£15.82	£0.86	£16.68	+£1.42	4 wk
Taiwan	£13.65	£0.93	£1.76	£0.17	£16.51	£0.61	£17.12	+£1.86	3 wk
Mexico	£14.86	£0.80	£1.16	£0.16	£16.98	£0.63	£17.61	+£2.35	3 wk
South Korea	£13.95	£1.24	£2.72	£0.21	£18.12	£0.00	£18.12	+£2.86	3 wk
Czech Republic	£15.92	£1.04	£1.62	£0.22	£18.80	£0.00	£18.80	+£3.54	2 wk
Poland	£16.07	£0.94	£1.58	£0.21	£18.80	£0.00	£18.80	+£3.54	2 wk
Germany	£17.89	£2.14	£4.17	£0.33	£24.53	£0.00	£24.53	+£9.27	2 wk
USA	£17.44	£2.02	£3.95	£0.33	£23.74	£0.86	£24.60	+£9.34	2 wk
United Kingdom	£18.50	£2.49	£3.75	£0.37	£25.11	£0.00	£25.11	+£9.85	1.5 wk
Japan	£18.19	£3.53	£3.60	£0.35	£25.67	£0.00	£25.67	+£10.41	3 wk

"Other" is energy, ESD packaging and cost of quality (rework and scrap at the country's defect rate). Components follow each country's sourcing index; labour, energy and duty are each country's own.

§6 — One-Time Automotive NRE

£16,200.00 · not in the unit cost

Item	One-time	Per board over one year (250,000)
PPAP	£2,600.00	£0.01
FMEA	£4,400.00	£0.02
DV / PV (DVP&R)	£7,000.00	£0.03
ASIL audit	£2,200.00	£0.01
Total NRE	£16,200.00	£0.06

What drives this board's cost

3 lines carry 56% of the board: Image sensor £4.64; LI-ION AND LI-POL CHARGER IC (BQ24025DRCR) £2.10; FPD-Link III Serializer With CSI-2 Interface (DS90UB935-Q1) £1.87. These are the quotes to get first.

Components are 87% of the board; bare board 3%, assembly and test 5%. A should-cost discussion on this board is a component-price discussion.

China (Shenzhen / Suzhou) is the lowest delivered cost of the 14 countries compared.

1 line of £1.00+ (£4.64) is priced from a range, not a quote: Image sensor.

Not in this unit cost

One-time automotive NRE (£16,200.00 — PPAP, FMEA, DV/PV, audit) is not in the unit cost; see §6.

Stencils, test fixtures and programming are inside the EMS price, spread over the order (no separate tooling line).

Tier-1 / distributor margin above the EMS price, and the enclosure, lens and cable of a module, are not in this board cost.

Analysis checks (1)

BOM taken from your BOM image (transcribed by the reader, never priced by it): 11 lines, 80 SMT placements. The photos were used for the board build and to fill gaps, not to write the BOM.

Limitations stated by the photo reader

The image sensor model is unknown

Source Photographs

2 images the costing was built from

The parts list came from the BOM image supplied with the analysis; the photographs gave the board's size, layer and assembly reading and any package the list left open.